CROWD-SOURCED TRUTH

A field guide for community data collection in hostile environments

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"The distance between what is said and what is known to be true has become an abyss."

Of all the things at risk, the loss of an objective reality is perhaps the most dangerous.

Mon Mothma - Andor

When The Official Story Doesn't Match The Ground Truth

Communities have been building their own evidence base for two decades. The pattern is consistent: when institutions fail to document what's happening, ordinary people step in.

Kenya 2008	45,000 SMS reports of election violence mapped in 6 days
Fukushima 2011	150M+ citizen radiation readings vs. single government averages
Syria 2012-present	650K+ videos preserved after YouTube deleted 200K
Ukraine 2022-present	6.9M+ videos archived; invasion detected via Google Maps traffic

The Three Pillars

Every crowd-sourced effort lives or dies on these axes

SPEED

How fast can you get useful data flowing?

SAFETY

How do you protect contributors & organizers?

RESILIENCY

Can the effort survive attacks and attrition?

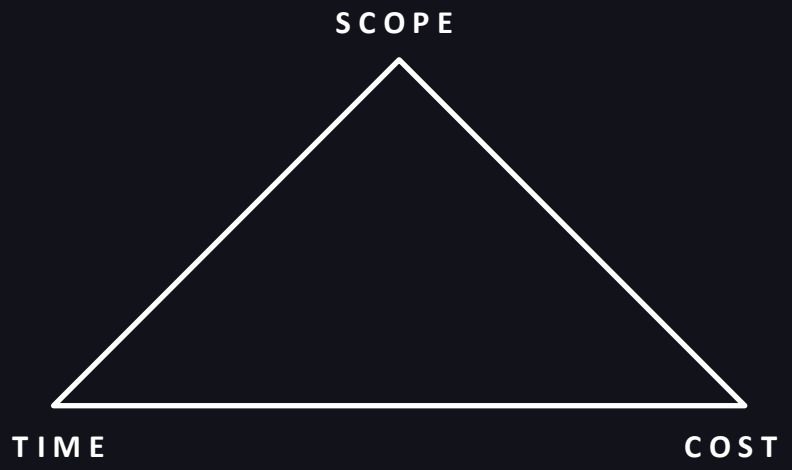
These pillars exist in productive tension. Speed demands simplicity; safety demands friction. The tradeoffs are features, not bugs.

Familiar Tradeoffs, Different Stakes

You already think in triangles. The skills transfer. The stakes don't.

The Scope Triangle

SDLC / project management

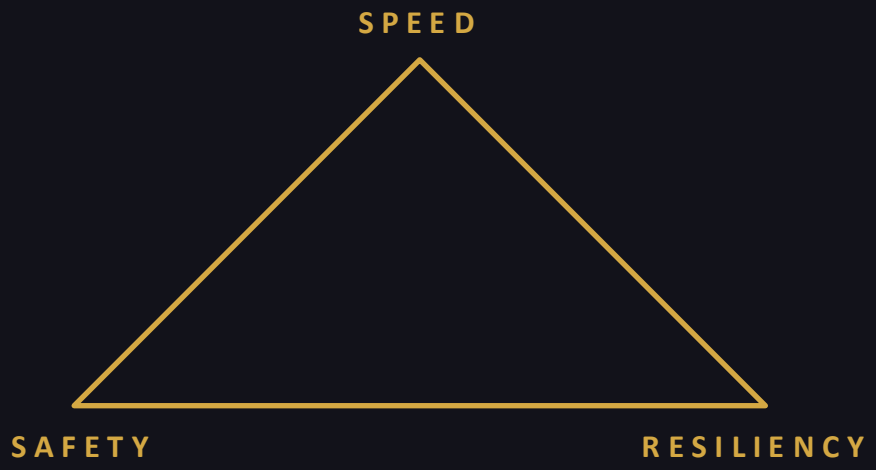


Pick two

Optimizes for: project delivery

The Crowd Source Triangle

community data under pressure



Navigate all three

Optimizes for: human outcomes & system survival

First: Is Someone Already Doing This?

Duplication wastes scarce volunteer energy and fragments the evidence base.

Who else is collecting similar data? Check local mutual aid networks, national coalitions, and international platforms.

Can you contribute to an existing effort? Consolidation strengthens the data.

How will you coordinate with adjacent efforts to avoid conflicts and share learning?

Who are the domain experts and legal advocates already in the space? Build relationships before you need them.

What Question Are You Answering?

Vague mandates produce unusable data

"Document everything"

"Track all the activity"

"Collect reports"

Specific questions produce actionable data

"What is the radiation on this street?"

"Can this specific claim be verified?"

"Where were checkpoints active today?"

Define the minimum viable data point: what fields, what geography, what time window.

The data you don't collect can't be weaponized.

Trust Without Identity

How do you build trust with contributors who need to stay anonymous?

Open Crowdsourcing	Anyone can submit. Trust through aggregation: clustering of reports from independent sources indicates reliability.	*Ushahidi, early crisis mapping*
Closed Crowdsourcing	Trained, vetted monitors with verified private identities but operational pseudonyms.	*Burundi's Amatoro mu Mahoro (450 monitors)*
Hybrid Verification	Open submission, but content must pass geolocation and chronolocation checks before publication.	*Bellingcat, CIR Ukraine Monitor Map*
Tiered Publication	Raw reports in one layer, verified reports in another. Users choose their confidence threshold.	*Modern crisis platforms*

Know Your Biases

Geographic Bias	Social media attention concentrates on populous, connected areas. Rural and infrastructure-poor regions are underrepresented.
Demographic Bias	Those with connectivity, language skills, and digital literacy dominate. The most vulnerable are least likely to report.
Survivorship Bias	You only hear from those who survived, escaped, or had access. Silenced voices shape the data by their absence.
Reporting Bias	Dramatic events get reported; slow-moving patterns don't. Frequency of reports does not equal severity.
Platform Bias	Your submission channel determines who can participate. SMS reaches different people than a web form or app.

Detecting Bad Faith And Data Poisoning

Bot Flooding	Automated submissions that overwhelm legitimate reports	Rate limiting, behavioral fingerprinting, CAPTCHAs for web forms
False Reports	Deliberate misinformation designed to discredit the dataset	Multi-source corroboration, geographic clustering analysis
Weaponized Flagging	Using platform rules to remove legitimate content	Independent archives, don't rely solely on third-party platforms
Trust Gaming	Bots simulate normal behavior to gain trust, then inject false data	Redundancy: multiple independent observers per event
Ai-generated Content	Synthetic text/images polluting crowdsourced surveys	Content authentication, metadata verification, human review layers

Protecting Contributors And Organizers

People have died when civic data projects failed at safety.

Who are your adversaries and what are their capabilities? State actors? Corporate? Vigilantes?

What is the MINIMUM data you need? Every unnecessary field is a potential weapon.

Strip metadata at point of ingestion, not after storage. Reduce precision to what your analysis actually needs.

What happens when someone is compromised? Compartmentalize knowledge. No single person's compromise should expose the entire operation.

Pre-establish secure backup communication channels that don't depend on potentially compromised infrastructure.

The Mosaic Effect

Individually harmless data points become dangerous when combined.

87%

of Americans uniquely identifiable from just 3 data points: ZIP + birthdate + gender

Real-world examples:

Strava heatmap exposed military bases, patrol routes, and 18,599 French military personnel across 100 bases

NYC taxi data was reverse-engineered to identify Muslim drivers via prayer time correlations

Netflix Prize data de-anonymized through correlation with public IMDb reviews

Design for the assumption that your adversary has access to datasets you don't know about.

Control Your Infrastructure

"We are writing our memories not in our own book but in a third party's book. We don't have control of it."

Hadi al Khatib, Syrian Archive founder, after YouTube deleted 200,000 videos of the Syrian conflict

Does your data live on infrastructure you control? Plan for the day platforms delete it.

How distributed is your architecture? Mirrors across jurisdictions resist legal takedowns.

What is your plan for internet shutdowns? (Egypt, Iran, Belarus, India have all imposed them)

How do you handle platform rules being weaponized against you?

"Too distributed to fail" = data replicated across enough independent nodes that no single action can remove it.

What To Do With The Data

Who benefits from this data?	CARE Principles: Collective Benefit, Authority to Control, Responsibility, Ethics
Who controls access?	Define roles, permissions, and revocation processes before collection begins
What are your retention policies?	How long do you keep raw data? When does it get aggregated or destroyed?
Could this data harm the people it describes?	Even well-intentioned collection can endanger vulnerable communities
Is this data creating legal obligations?	Evidence preservation, data protection laws, cross-border data transfer rules
How do you publish responsibly?	Aggregate, anonymize, delay. The data you release cannot be un-released.

Plan Your Exit Before You Start

Sustainability is the silent killer of civic tech. Plan for three scenarios:

Orderly Wind-down	Mission accomplished or funding exhausted. Who inherits the data? How is community trust transferred?
Forced Shutdown	Legal action, platform takedown, or political pressure. Where does the data go? What gets preserved vs. destroyed?
Organizer Compromise	Key people arrested, surveilled, or turned. Who has backup credentials? What's the succession plan?

When It Goes Wrong

Boston Marathon Reddit	Crowdsourced investigations lack due process. Social media turns speculation into viral accusation.
Syria: YouTube deletions	200K+ videos destroyed by automated content moderation. Evidence must live on infrastructure you control.
Tunisia: ISP code injection	Government stole activist passwords through the ISP itself. The network layer is adversarial.
Haiti: SMS pleas published	Sensitive personal location data made public without consent. Good intentions don't prevent harm.
Bridgefy app (HK/Belarus)	Speed of adoption outran security vetting. Researchers could read messages and build social graphs.

Tensions Cannot Be Resolved, Only Managed

Speed	vs	**Rigor**	How fast do you need data vs. how sure do you need to be?
Openness	vs	**Security**	Open systems support verification but also enable adversaries.
Documentation	vs	**Danger**	Every data point that documents an abuse potentially identifies a victim.
Distribution	vs	**Governance**	Distributed systems resist censorship but are harder to manage.

Your Planning Checklist

- What specific question are we answering?
- Does someone already collect this? Can we join them?
- What is the minimum data we need?
- What verification model fits our speed and consequence?
- What biases will our method bake in?
- How do we detect and filter bad-faith submissions?
- How do we protect contributor anonymity against the mosaic effect?
- Do we control our own infrastructure?
- What happens when someone is compromised?
- What are our obligations after collection?
- What is our exit plan?

Take It With You

What you'll find:

- Full field guide document (this talk + more)
- Planning checklist (printable)
- Curated list of open-source repos and tools
- People and organizations to follow
- Case study reading list
- Academic references on verification & bias

Scan for resources

Remember: Crowd-sourced civic data has contributed to ICC proceedings, criminal convictions in Germany, Sweden, Finland, and the Netherlands, and international sanctions regimes. The Berkeley Protocol establishes that digital open-source evidence meets courtroom standards. This work has real consequences.

"I have friends everywhere."

Thank you.

Sources & Citations (1/2)

Standards & Protocols

Berkeley Protocol on Digital Open Source Investigations (2020)	ohchr.org/...berkeley-protocol
CARE Principles for Indigenous Data Governance (Carroll et al., 2020)	datascience.codata.org/articles/dsj-2020-043
ICRC: The Mosaic Effect - Revelation Risks of Combining Data	blogs.icrc.org/.../mosaic-effect-revelation-risks

Key Case Studies

Ushahidi: From Crisis Mapping Kenya to Mapping the Globe	tavaana.org/.../ushahidi-crisis-mapping
Bellingcat: OSINT and War Crime Accountability in Ukraine	rusi.org/.../osint-war-crime-accountability-ukraine
HRW: Social Media Platforms Remove Evidence of War Crimes	hrw.org/.../video-unavailable
Syrian Archive / Mnemonic: Digital Evidence Preservation	mnemonic.org
Safecast: Citizen Radiation Monitoring	safecast.org

Additional sources referenced in this talk: Sweeney (2000) on re-identification; Freedom House Freedom on the Net reports; Strava heatmap investigations (Le Monde, 2025); Fang et al. & Miao et al. on adversarial crowdsourcing; Wiggins et al. (2011) on citizen science validation; COINTELPRO documentation; Hargittai (2020) on big data bias. Full bibliography in the downloadable resource guide.

All URLs verified April 2026. Full resource guide with additional links available via QR code.

Sources & Citations (2/2)

Academic Research

Starbird et al.: Misinformation in Crisis (UW)	medium.com/@katestarbird
Crowdsourcing Under Data Poisoning Attacks (Springer)	link.springer.com/.../978-3-030-49669-2_18
Spatial Biases in Crowdsourced Data (ScienceDirect)	sciencedirect.com/.../S0198971520302477
Collective InfoSec in Hong Kong Protests (USENIX)	usenix.org/.../albrecht

Tools & Resources

Bellingcat Online Investigation Toolkit	bellingcat.gitbook.io/toolkit
awesome-osint: Curated OSINT Tools List	github.com/jivoi/awesome-osint
Ushahidi Platform (GitHub)	github.com/ushahidi
SecureDrop: Whistleblower Submission System	securedrop.org
EFF Surveillance Self-Defense Guide	ssd.eff.org

All URLs verified April 2026. Full resource guide with additional links available via QR code.